ASSIGNMENT-17.3

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BATCH-41

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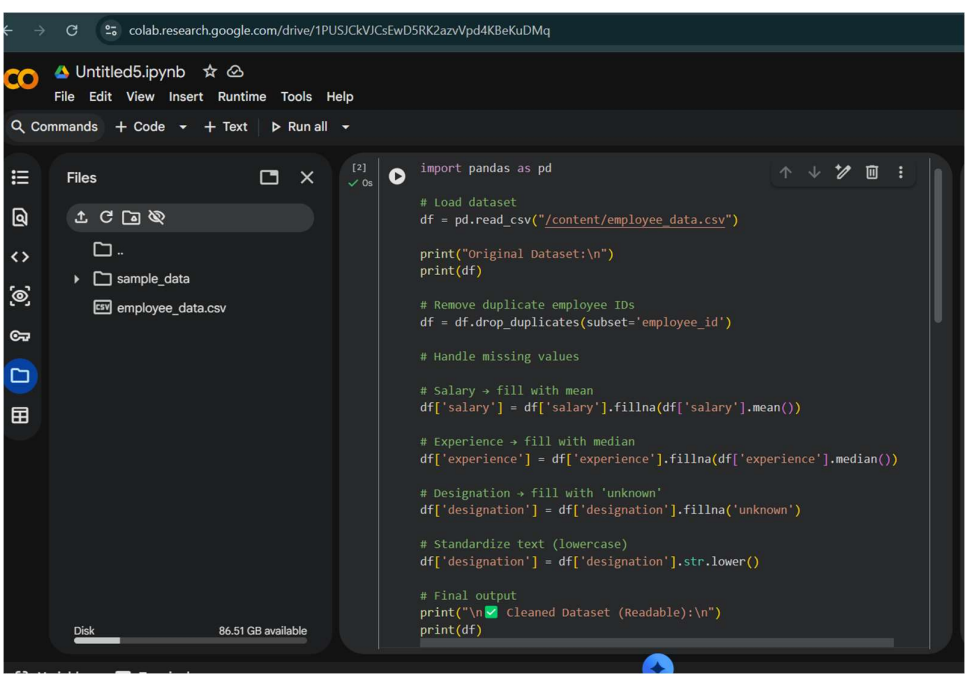
TASK 1: Employee Dataset Cleaning

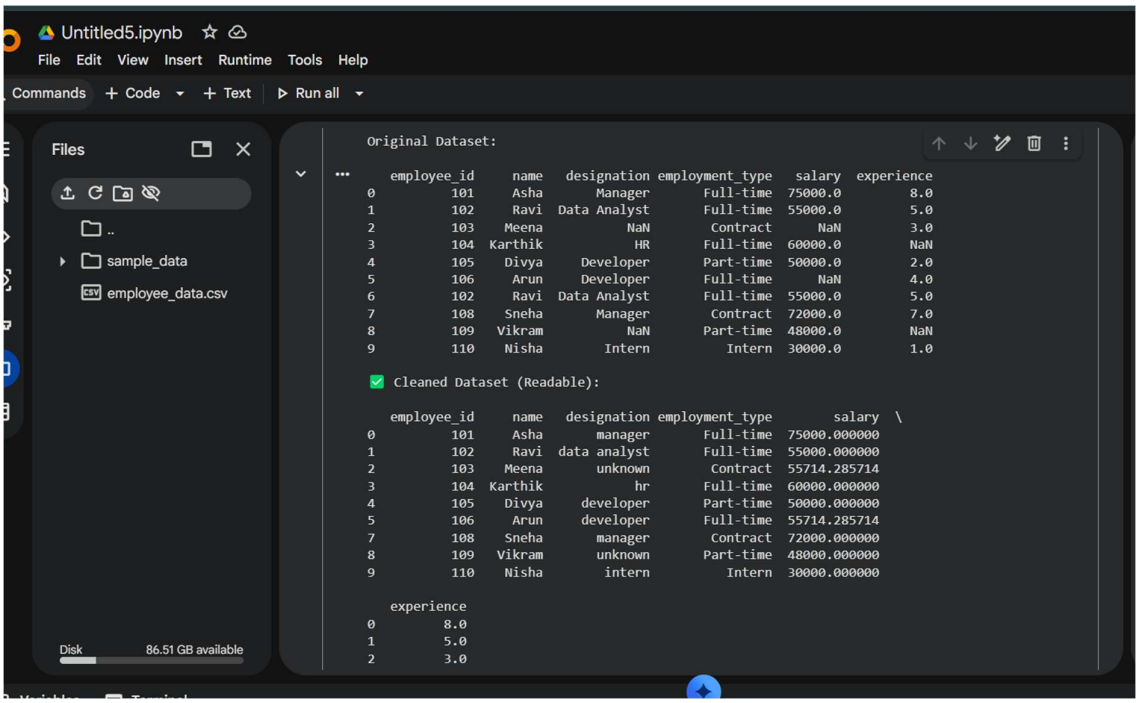
PROMPT:

Generate a Python script using pandas to clean and preprocess an employee dataset. The dataset contains missing values, duplicate records, and inconsistent text formatting. Handle missing values in salary and experience by replacing them with appropriate statistical measures such as mean and median, and fill missing designation values with "unknown". Remove duplicate employee IDs to ensure data uniqueness. Standardize text fields by converting all job titles to lowercase. Finally, display the cleaned dataset in a readable format suitable for further analysis and machine learning tasks.

CODE:



OUTPUT:



JUSTIFICATION:

Data cleaning is important because raw data often contains errors like missing values and duplicates. Handling missing salary and experience ensures the dataset remains complete and usable. Removing duplicate employee IDs avoids incorrect analysis and repetition. Standardizing text (like job titles) helps maintain consistency in the data. Filling unknown values prevents loss of important records. Overall, preprocessing makes the dataset clean, reliable, and ready for accurate analysis or machine learning.

TASK-2: Student Performance Data Preprocessing

PROMPT:

Generate a Python script using pandas to preprocess student performance data. Convert the exam_date column into datetime format and extract features such as month, day, and semester. Normalize the total_marks column using Min-Max scaling. Detect outliers in marks using the IQR method and handle them appropriately. Finally, display both the original and transformed datasets for comparison.

CODE:

Untitled5.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Files

..

sample_data

employee_data.csv

student_data.csv

Disk 86.51 GB available

[4]

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
# Load dataset
df = pd.read_csv("/content/student_data.csv")
print("Original Dataset:\n")
print(df)
# Convert exam_date to datetime
df['exam_date'] = pd.to_datetime(df['exam_date'])
# Extract features
df['month'] = df['exam_date'].dt.month
df['day'] = df['exam_date'].dt.day
# Semester logic
df['semester'] = df['month'].apply(lambda x: 1 if x <= 6 else 2)
# Detect outliers using IQR
Q1 = df['total_marks'].quantile(0.25)
Q3 = df['total_marks'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
# Replace outliers with median
median = df['total_marks'].median()
df['total_marks'] = df['total_marks'].apply(
    lambda x: median if x < lower_bound or x > upper_bound else x
)
# Normalize marks
scaler = MinMaxScaler()
df['normalized_marks'] = scaler.fit_transform(df[['total_marks']])
print("\n✅ Transformed Dataset:\n")
print(df)
```

OUTPUT:

```
Original Dataset:

   student_id     name   exam_date  total_marks  subject
0         201     Asha  2024-01-15          450     Math
1         202     Ravi  2024-02-20          480  Science
2         203    Meena  2024-03-10          300  English
3         204  Karthik  2024-04-05          900     Math
4         205    Divya  2024-05-18          420  Science
5         206     Arun  2024-06-25           50  English
6         207    Sneha  2024-07-12          470     Math
7         208   Vikram  2024-08-30          490  Science
8         209    Nisha  2024-09-14         1000  English
9         210    Rahul  2024-10-22          430     Math

✅ Transformed Dataset:

   student_id     name  exam_date  total_marks  subject  month  day  semester  \
0         201     Asha 2024-01-15        450.0     Math      1   15         1
1         202     Ravi 2024-02-20        480.0  Science      2   20         1
2         203    Meena 2024-03-10        460.0  English      3   10         1
3         204  Karthik 2024-04-05        460.0     Math      4    5         1
4         205    Divya 2024-05-18        420.0  Science      5   18         1
5         206     Arun 2024-06-25        460.0  English      6   25         1
6         207    Sneha 2024-07-12        470.0     Math      7   12         2
7         208   Vikram 2024-08-30        490.0  Science      8   30         2
8         209    Nisha 2024-09-14        460.0  English      9   14         2
9         210    Rahul 2024-10-22        430.0     Math     10   22         2

   normalized_marks
```

```
2         203    Meena 2024-03-10        460.0  English      3   10         1
3         204  Karthik 2024-04-05        460.0     Math      4    5         1
4         205    Divya 2024-05-18        420.0  Science      5   18         2
5         206     Arun 2024-06-25        460.0  English      6   25         2
6         207    Sneha 2024-07-12        470.0     Math      7   12         2
7         208   Vikram 2024-08-30        490.0  Science      8   30         2
8         209    Nisha 2024-09-14        460.0  English      9   14         2
9         210    Rahul 2024-10-22        430.0     Math     10   22         2

   normalized_marks
0          0.428571
1          0.857143
2          0.571429
3          0.571429
4          0.000000
5          0.571429
6          0.714286
7          1.000000
8          0.571429
9          0.142857
```

JUSTIFICATION:

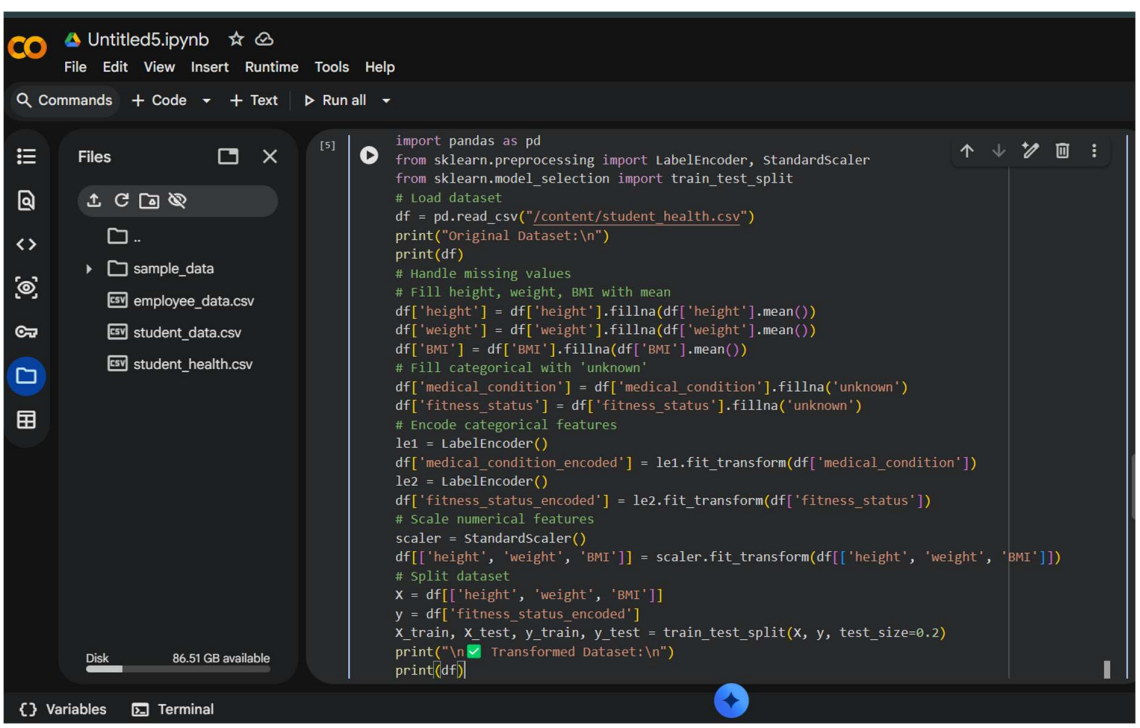
Data preprocessing helps improve the quality of student performance data. Converting exam dates into datetime format allows extraction of useful features like month and semester. Detecting outliers ensures extreme values do not affect analysis. Replacing outliers with median keeps the data balanced. Normalizing marks brings all values into a common scale for better comparison. Overall, preprocessing makes the dataset more meaningful and suitable for analysis or machine learning.

TASK-3: Student Health Data Preparation

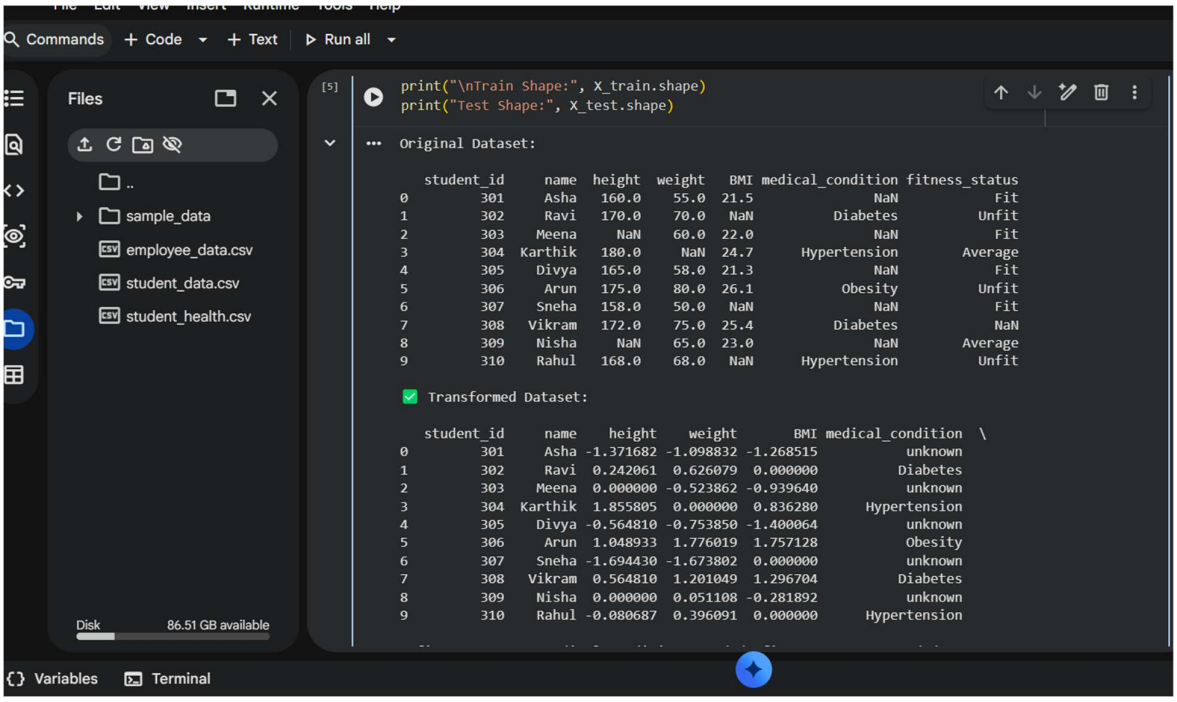
PROMPT:

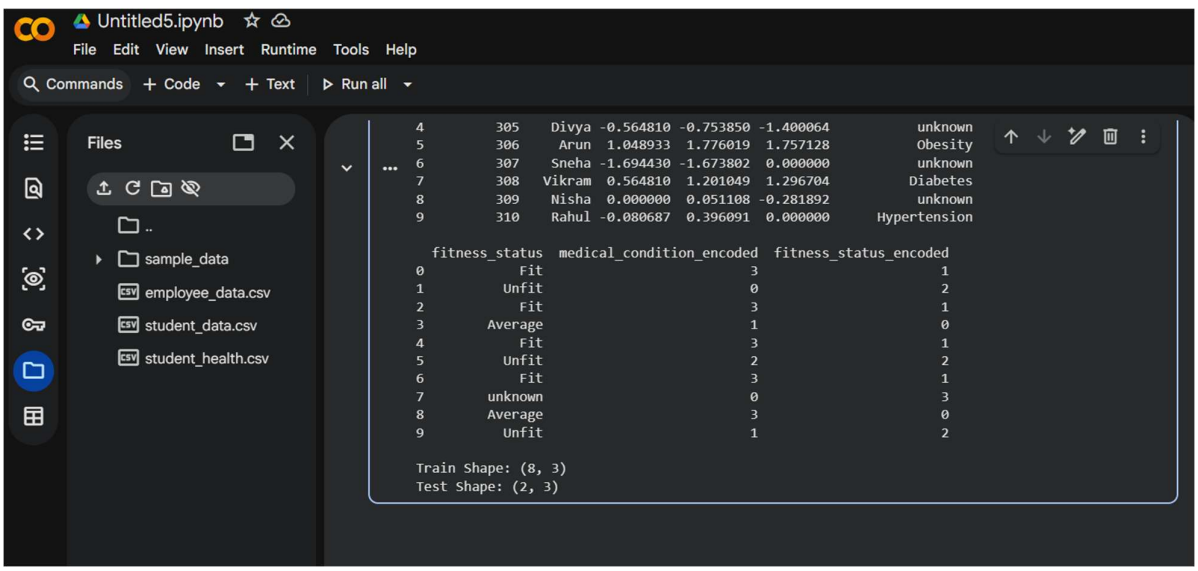
Generate a Python script using pandas and sklearn to preprocess student health data. Handle missing values in height, weight, and BMI using appropriate statistical methods. Encode categorical features such as medical_condition and fitness_status. Apply standard scaling to numerical features. Finally, split the dataset into training and testing sets and display the processed output.

CODE:



OUTPUT:





JUSTIFICATION:

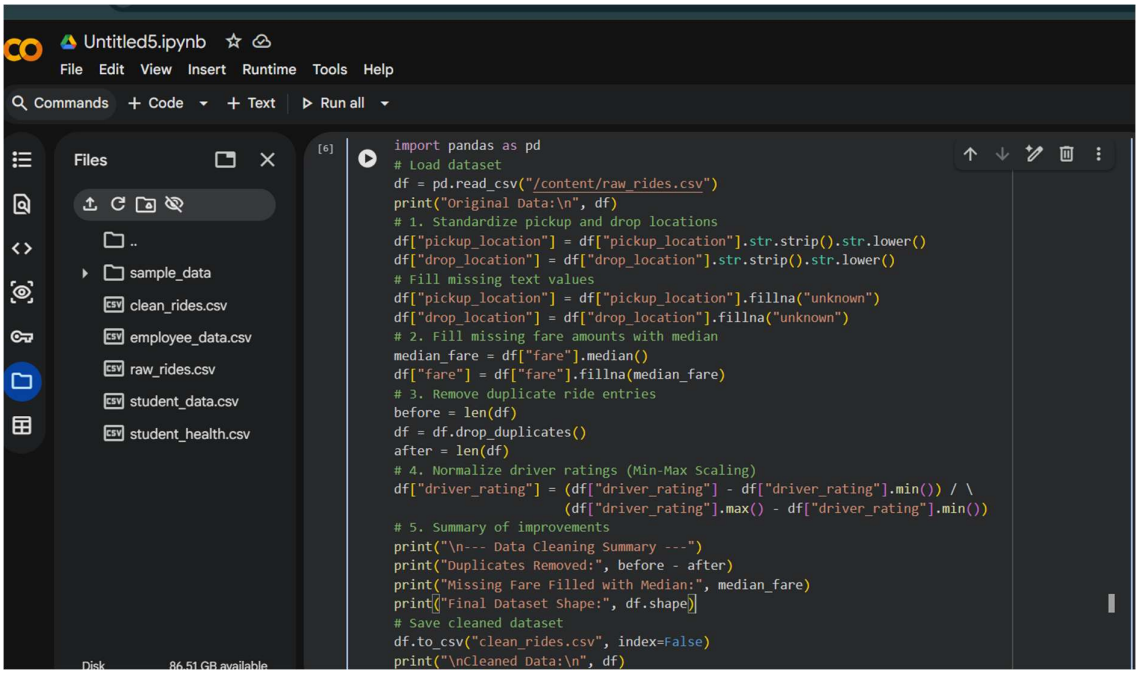
Data preprocessing is important to make health data accurate and usable. Missing values in height, weight, and BMI are filled to avoid incomplete records. Encoding categorical data converts text into numbers for machine learning models. Standard scaling ensures all numerical values are on the same scale. Splitting data into training and testing helps evaluate model performance. Overall, preprocessing improves data quality and supports better health predictions.

TASK-4: Real-Time Application: Ride-Sharing Data Cleaning

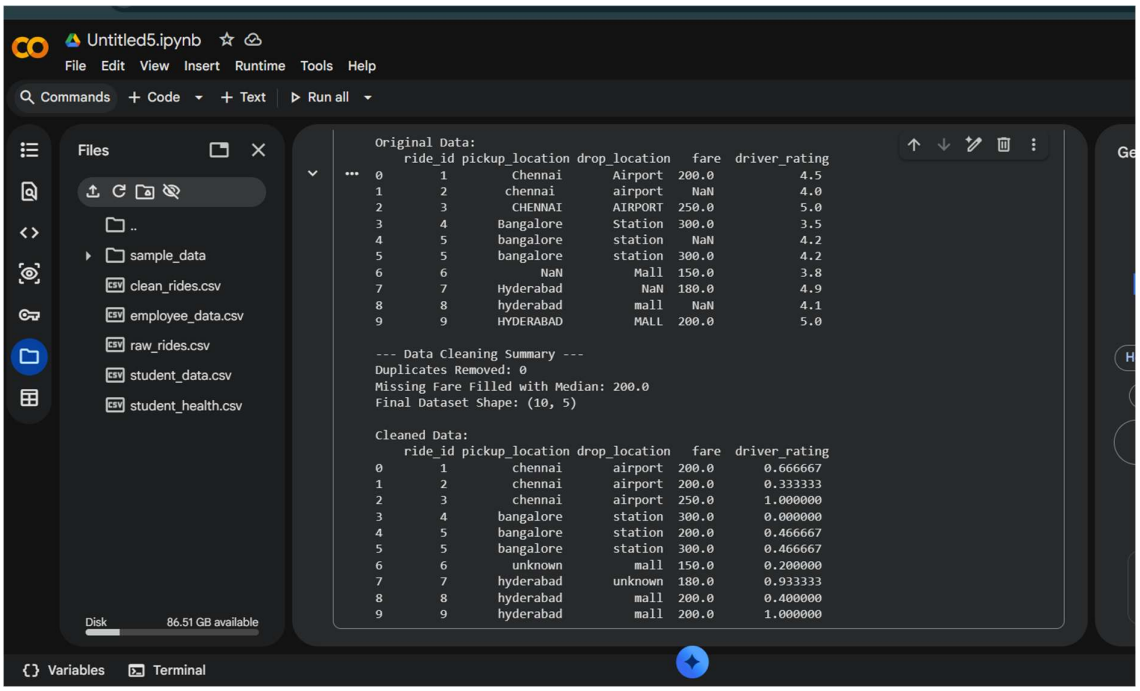
PROMPT:

Generate a Python script using pandas to clean ride-sharing data. Standardize pickup and drop location names by converting them into a consistent format. Handle missing fare amounts by filling them with the median value. Remove duplicate ride entries. Normalize driver ratings using Min-Max scaling. Finally, display the cleaned dataset and provide a summary of improvements made during preprocessing.

CODE:



OUTPUT:



JUSTIFICATION:

Data cleaning is essential for improving the quality of ride-sharing data. Standardizing location names ensures consistency in analysis. Missing fare values are filled using the median to avoid bias from extreme values. Duplicate ride entries are removed to prevent incorrect calculations. Driver ratings are normalized to bring them into a common scale. Overall, preprocessing makes the dataset accurate and ready for operational analysis.